

Dose Exploration Based on Quantitative Pharmacokinetic Analysis of Monoclonal Antibodies for Pediatric Immune-Mediated Diseases: Current Evidence and Future Directions

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Abstract: Monoclonal antibodies have greatly changed the treatment of pediatric immune-mediated diseases (PIMDs) and have become the main treatment method for patients who do not respond to conventional therapy. However, there are many challenges in developing pediatric dosing regimens. The pharmacokinetics of monoclonal antibodies differ between adults and children due to dynamic, age-related physiological changes that affect target-mediated clearance, FcRn-mediated recycling, and fluid volume distribution. Pediatric dosing regimens typically employ weight-based and weight-stratified dosing that are extrapolated based on adult data and often result in subsequent treatment failure. This review provides a comprehensive analysis of current research on the pharmacokinetics of pediatric monoclonal antibodies, discussing the application of population pharmacokinetics and physiologically based pharmacokinetic modeling. In addition, the exposure-response relationship and the role of therapeutic drug monitoring in dose optimization are summarized. Overall, the current understanding of pharmacokinetics and pharmacodynamic behaviors specific to the child population is not comprehensive, and clinical translation remains hindered by the diversity of reported endpoints and the lack of external validation for existing models. Therefore, the future of precision dosing lies in moving beyond simple exposure matching to pharmacodynamic (PD) endpoint-based treatment regimens. Finally, we propose a feasible pathway that integrates next-generation mechanistic models with artificial intelligence (AI) and clinical decision support systems (CDSS) to bridge the gap between computational evidence and clinical applications.

Keywords: biologics, children, pharmacometrics, precision medicine, therapeutic drug monitoring

Introduction

Monoclonal antibodies (mAbs) play a critical role in PIMDs. Specifically, they serve as first-line biologics for patients unresponsive to conventional synthetic therapies.^{1,2} In recent years, with advancing research, the US Food and Drug Administration (FDA) and the European Medicines Agency (EMA) have approved a growing number of mAbs for pediatric indications such as pediatric inflammatory bowel disease (PIBD), juvenile idiopathic arthritis (JIA) and uveitis, which has improved clinical outcomes for affected children.³⁻⁵ However, these drugs in pediatric populations still face unique challenges, with efficacy and safety highly dependent on the design of dosing regimens. For example, the standardized weight-based dosing regimen widely used in pediatric clinical practice is being challenged by recent research, with up to 40% of patients experiencing primary or secondary treatment failure.³ Real-world data show that only 46.9% of pediatric patients

reached target trough concentrations (C_{trough}) with standard infliximab dosing regimens, and over 20% required treatment optimization during induction due to subtherapeutic concentrations.^{4,5} One primary reason for this prevalent suboptimal clinical response is inadequate drug exposure, stemming from the wide between-patient variability in drug pharmacokinetics.^{5,6}

mAbs are not affected by traditional hepatic metabolism or renal excretion. Instead, their pharmacokinetics are driven by three mechanisms: non-specific cellular uptake, particularly fluid-phase pinocytosis by vascular endothelial cells, followed by lysosomal proteolytic degradation, protection from degradation through neonatal Fc receptor (FcRn)-mediated recycling, and target-mediated drug disposition (TMDD).^{7,8} In the pediatric population, these processes are heavily influenced by age-related dynamic developmental changes (Figure 1).⁹ For example, age-related changes in plasma volume and extracellular fluid can alter the distribution of mAbs, while the maturation of endothelial function and FcRn expression directly affect the clearance.⁶ Additionally, for subcutaneous administration, developmental differences in lymphatic drainage and subcutaneous adipose tissue can alter the absorption of mAbs in children.^{10,11} Furthermore, changes in target receptor expression and inflammatory burden at different developmental stages can significantly modify TMDD, resulting in highly non-linear and unpredictable exposure of mAbs in children under standard dosing regimens.⁵

Beyond these mechanistic complexities, a major practical challenge remains that pediatric drug development heavily relies on extrapolation from adult cohorts due to a severe shortage of high-quality pediatric PK data.^{12,13} This regulatory approach has been recently formalized by the ICH E11A “Pediatric Extrapolation” guideline, finalized in December 2024, designed to maximize the use of adult data. Extrapolation relies on the fundamental assumption that the disease, drug pharmacology, and clinical response to treatment are sufficiently similar between the adult and intended

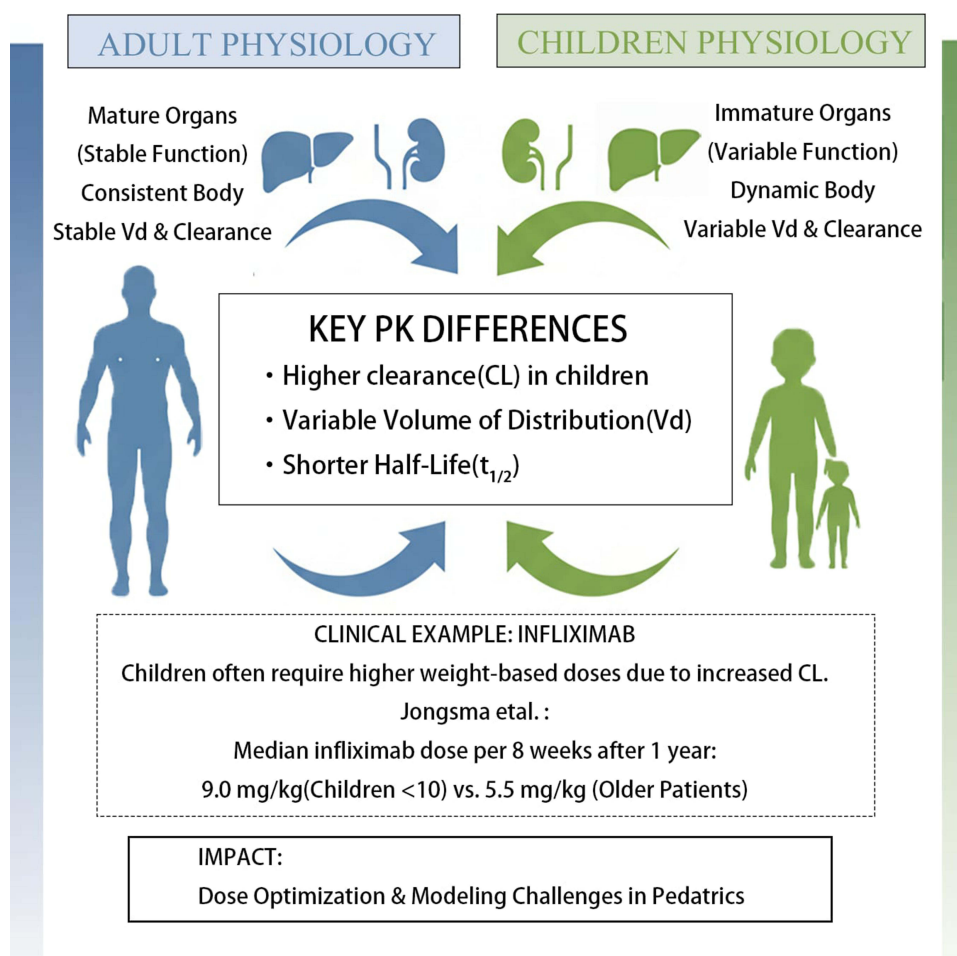


Figure 1 Overview of Differences in Monoclonal Antibody Pharmacokinetics Between Pediatric and Adult Patients.

pediatric populations.¹³ For PIMDs treated with mAbs, these extrapolation criteria are generally met. Pediatric and adult patients share similar pathophysiological inflammatory cascades, and mAbs maintain consistent target engagement and pharmacodynamic (PD) profiles across age groups.^{5,6}

Consequently, achieving comparable drug exposure provides a scientific basis for assuming comparable clinical response.¹³ To scientifically justify this extrapolation and account for age-dependent PK variations, the ICH E11A guideline explicitly identifies quantitative modeling and simulation (M&S) as essential tools. M&S bridges these qualitative similarities into actionable pediatric dosing regimens through exposure matching.

Collectively, this review aims to examine the applications of quantitative PK/PD modeling for mAbs in PIMDs. Specifically, this review addresses the following 2 topics: (1) the use of population pharmacokinetics (PopPK) and physiologically based pharmacokinetics (PBPK) models for pediatric extrapolation and pre-approval dose selection; and (2) the implementation of model-informed Therapeutic Drug Monitoring (TDM) to optimize individualized dosing regimens in post-approval clinical practice. Ultimately, the review explores feasible pathways to improve model interpretability, predictive accuracy, and universality. It provides future research directions for achieving precise pediatric monoclonal antibody therapy. The detailed literature search strategy and study selection criteria utilized for this review are provided in [Table S1](#) and [Figure S1](#).

Quantitative Modeling for Pediatric Extrapolation and Dose Selection

Development of Population-Based and Physiologically Based Pharmacokinetic Models

Pediatric patients should not be considered as small adults. As physiological parameters change quickly with age, adult dosing regimens are not only ineffective but also potentially unsafe.^{6,9} To achieve individualized dosing, researchers mainly employ two complementary PK modeling strategies. The top-down PopPK approach utilizes advanced statistical models to analyze real-world clinical data, identifying specific covariates, such as developmental physiology and disease status, to explain individual variability in drug exposure and clearance. However, its predictive power remains largely constrained by the specific covariates captured within a given study.¹⁴ In contrast, bottom-up PBPK approaches integrate physiological, anatomical, and drug-specific data to predict drug distribution, even before large-scale clinical trials are launched. Yet, their application is often limited by a reliance on extensive pre-existing mechanistic parameters.^{5,15} Furthermore, a critical limitation in purely pharmacokinetic predictions is that clinical responses are usually not binary (ie, “effective” or “ineffective”), and there is not a single effective dose for every rheumatic disease. To bridge this gap, PK/PD modeling serves as a quantitative tool to link PK concentration curves to therapeutic efficacy and safety indicators.¹⁶ This comprehensive approach forms the core of model-informed drug development (MIDD) strongly advocated by regulatory agencies. Notably, the US FDA’s guidance on “General Clinical Pharmacology Considerations for Pediatric Studies” explicitly emphasizes the utilization of M&S. These quantitative tools are recommended to characterize the impact of developmental changes on PK/PD and to support dose selection across pediatric age groups.¹² Overall, while each modeling approach possesses distinct methodological limitations, this regulatory framework highlights the necessity of a multifaceted evidence foundation to establish optimal dosing regimens. By combining these complementary methods, researchers can build an integrated basis for mAbs treatments in children.

The core task in developing PopPK models is identifying the influence of key covariates on critical PK parameters across multiple monoclonal antibody studies ([Table 1](#)). Body weight has been identified as the primary determinant of PK variability.⁹ This modeling consensus directly supports weight-based or weight-stratified dosing regimens, which have been successfully validated in recent pediatric clinical trials.¹⁷ Beyond simple anthropometric measures, disease-related factors are also identified as critical covariates. One observational study suggested that compared to those with idiopathic uveitis, in children with JIA, the clearance of adalimumab was 52% higher.¹⁸ Underlying this disease variability is often the inflammatory burden. Studies on infliximab and golimumab demonstrated that markers, such as lower baseline serum albumin and higher erythrocyte sedimentation rate (ESR), correlated with increased drug clearance.^{19,20} Similarly, an updated emapalumab model revealed that drug clearance is proportionally dependent on total interferon-gamma concentrations, highlighting the need to adjust dosing based on specific inflammatory states.²¹ Although PopPK modeling is limited by the specific covariates captured during a given study, the modeling framework for mAbs in children has

Table 1 Representative PopPK and PBPK Modeling Studies for Pediatric Monoclonal Antibodies

Modeling Approach	mAb	Target Population/ Disease	Main Application	Ref
PopPK	Infliximab & Golimumab	Pediatric IBD / UC	Incorporated body weight as the primary size descriptor via allometric scaling, and identified lower baseline serum albumin and higher ESR as covariates correlating with increased clearance.	Bauman 2020, ¹⁹ Xu 2019 ²⁰
PopPK	Adalimumab	JIA vs Idiopathic Uveitis	After adjusting for body weight, demonstrated a 52% higher clearance in children with JIA compared to idiopathic uveitis.	Golhen 2025 ¹⁸
PopPK	Emapalumab	sJIA/HLH (Inflammatory states)	In addition to standard body weight scaling, revealed that drug clearance is proportionally dependent on total interferon-gamma concentrations.	Brossard 2024 ²¹
PBPK	Adalimumab & Ustekinumab	JIA / Pediatric IBD	Scaled and validated adult PBPK models to pediatric populations to predict drug disposition.	Liu 2025, ²⁴ Cai 2024 ²⁵
PBPK	Anakinra	Critically ill children receiving ECLS	Incorporated CRRT/ECMO compartments and increased total body weight by 30% to account for edema, demonstrating renal dose adjustments are unnecessary.	Dubinsky 2024 ²⁶
PBPK	Infliximab	Pediatric IBD	Evaluated PK profiles when switching from intravenous to subcutaneous administration across different pediatric weight groups.	Petric 2024 ²⁷

Abbreviations: ECLS, Extracorporeal Life Support; ESR, Erythrocyte Sedimentation Rate; HLH, Hemophagocytic Lymphohistiocytosis; IBD, Inflammatory Bowel Disease; IFN-gamma, Interferon-gamma; IV, Intravenous; JIA, Juvenile Idiopathic Arthritis; mAb, Monoclonal Antibody; PBPK, Physiologically Based Pharmacokinetics; PK, Pharmacokinetics; PopPK, Population Pharmacokinetics; SC, Subcutaneous; sJIA, systemic Juvenile Idiopathic Arthritis; UC, Ulcerative Colitis.

been continuously refined through cross-cohort syntheses and model-based meta-analysis (MBMA).²² These integrated methods allow for the systematic incorporation of new variables specific to certain diseases, thereby gradually enhancing the predictive precision for the pediatric populations.^{15,23}

In contrast to PopPK approaches, PBPK modeling enables predictions by constructing virtual pediatric populations. Besides commercial platforms such as SimCYP[®], GastroPlus[®] and PK-Sim, general programming languages like R, Matlab and Julia are also widely used to build complex mAbs PBPK models due to their high flexibility in customization. A common strategy involves first developing and validating PBPK models in adults before scaling them to pediatric populations. It has been successfully applied to adalimumab and ustekinumab.^{24,25} The core advantage of PBPK modeling lies in its incorporation of a deeper mechanistic framework, which allows for extrapolation beyond the studied populations and evaluated doses.^{5,6} A case in point is predicting anakinra exposure in critically ill children, which requires modifying the PBPK model to incorporate specific compartments (eg, extracorporeal life support systems).²⁶ Furthermore, this versatility also enables models to explore hypothetical dosing regimens across different routes of administration. A key example is the evaluation of switching from intravenous to subcutaneous infliximab across different pediatric weight groups.²⁷

Furthermore, the frontier of PK modeling is advancing towards more personalized approaches. Beyond standard covariates, recent studies have incorporated genetic polymorphisms (eg, ADAM17 SNPs) and drug-drug interactions (eg, IVIG co-administration) to better explain individual variability.^{28,29} For novel biologics, model structures are also evolving. To describe the PK of the long-acting Fc-fusion protein HL2351, Ngo et al developed a complex model incorporating intricate mechanisms such as neonatal Fc receptor (FcRn)-mediated recycling and target binding.³⁰ At the same time, the development of these models relies on data from clinical trials in pediatric populations. The Phase 1 study by Mallalieu et al in children under two years old with systemic juvenile idiopathic arthritis (sJIA) provided critical PK and safety data for tocilizumab.³¹ These advanced models and valuable data represent a significant step towards more personalized predictions.

Model-Informed Extrapolation Across Age Cohorts

Model-informed extrapolation (MIE) is a key strategy to address lengthy traditional pediatric drug development cycles and non-standard off-label use.³² This approach is based on the principle that achieving drug exposure levels in children

equivalent to those in adults will result in similar efficacy. This principle has been supported across various biologic therapies.^{20,33}

To technically execute this extrapolation, the foundational step involves allometric scaling. Typically, clearance (CL) and volume of distribution (Vd) are scaled by body weight using standard fixed exponents.⁹ Notably, simple allometric scaling is often insufficient for mAbs, particularly in neonates and young infants. Therefore, age-dependent maturation functions are frequently incorporated to account for the developmental ontogeny of the non-specific clearance pathways (eg, fluid-phase pinocytosis in endothelial cells) and age-related shifts in interstitial fluid volumes. Additionally, these functions are essential to capture the dynamic changes in endogenous IgG levels, which compete for FcRn-mediated recycling and directly impact mAb clearance.^{6,19} Furthermore, beyond body weight and age, pediatric extrapolation for mAbs requires disease-specific considerations. Differences in inflammatory burdens, baseline target receptor expression, and soluble target shedding can alter nonlinear TMDD. This necessitates precise adjustments in model to accurately capture concentration-dependent clearance and prevent biased dose predictions.^{5,8,19}

For example, demonstrating the bidirectional utility of MIE, an analysis by Baldauf et al used Bayesian and popPK modeling for Canakinumab in sJIA and adult-onset Still's disease. Specifically, information was borrowed from five pooled sJIA studies. These data served as the informative prior using a robust meta-analytic predictive (MAP) approach. This prior was then updated with AOSD clinical data. The approach demonstrated that similar clinical outcomes were achieved when drug exposure was comparable across both populations, supporting a pediatric-to-adult reverse extrapolation.³⁴ Additionally, for pediatric systemic lupus erythematosus (pSLE), where it is difficult to recruit sufficient patients, the US FDA employed Bayesian statistics to support approval of intravenous belimumab in the US.³⁵ To achieve this, a post-hoc Bayesian analysis was conducted. Information was directly borrowed from adult Phase 3 studies (BLISS-52 and BLISS-76). These adult data formed a mixture prior which was then updated using the pediatric data from the PLUTO trial. The approach successfully estimated the treatment effect and justified the clinical extrapolation.³⁵ MIE maximizes the use of adult information, thereby meeting scientific and regulatory requirements to bring new medicines to children. Beyond supporting initial approval, M&S are instrumental in optimizing dosing frequency and formulations. For instance, regarding etanercept, researchers utilized PopPK modeling and Monte Carlo simulations to compare once-weekly (0.8 mg/kg) and twice-weekly (0.4 mg/kg) regimens. The results showed highly overlapping steady-state concentration profiles for both regimens. This simulation evidence directly supported US FDA approval of the more convenient once-weekly dosing regimen.³⁶

Simulating Target Engagement and Alternative Therapeutic Strategies

Beyond elucidating in vivo PK variations, exploration of the relationship between concentration and therapeutic effect is essential. PK/PD models not only describe drug distribution but also focus on quantifying the effects on relevant biomarkers or target cells. For example, Brossard et al developed a PK/PD model for emapalumab that directly quantified the extent to which drug concentrations inhibited inflammatory markers, estimating a 98.3% reduction in CXCL9 synthesis rate and a 99.6% reduction in ferritin.³⁷ Similarly, models have been used to describe the effect of anakinra on C-reactive protein (CRP) levels and canakinumab on total interleukin-1 β .^{38,39} This approach has also been extended to cellular targets, such as models describing the depletion effect of rituximab on CD19+ B cells. A study by Pan et al quantified that rituximab increased the mortality rate of CD19+ B cells by 35-fold, while another study by Riva et al used a turnover model to characterize CD19+ suppression.^{40,41} However, in different studies, the results of the model predicting the half-life of rituximab and CD19+ cells differed significantly due to differences in patient populations and research methods. The core value of the PK/PD model is the ability to simulate the expected therapeutic effect, enabling precise dose selection based on PD endpoints. Existing studies, however, have shown that when models are applied to different disease stages or specific pediatric disease cohort without external validation, prediction bias often occurs.^{42–44}

In the strategic design of mAb regimens, apart from dosage calculation, these quantitative simulation tools are crucial for exploring administration routes and formulation selection. For many mAbs, transitioning from intravenous infusion (IV) to subcutaneous injection (SC) offers clear convenience.¹⁰ However, optimizing SC regimens requires a deep understanding of the absorption. Because mAbs are primarily absorbed through the lymphatic system, age-dependent variations in subcutaneous tissue and lymphatic capillary maturation can lead to variable absorption rates in children.^{10,11} Real-world data and

simulation studies indicate that SC regimens for drugs like abatacept and infliximab can maintain or even improve C_{trough} levels compared to IV induction.^{45,46} However, M&S also indicated that fixed-dose subcutaneous administration may result in higher exposure in children and lower exposure in those with high BMI.¹¹ Therefore, quantitative modeling remains essential to further refine pediatric SC regimens and overcome this absorption variability.^{10,45,46}

Beyond dosage and route selection, quantitative evaluation extends to the clinical adoption of alternative drug formulations, such as biosimilars. Extensive real-world cohorts and randomized trials have validated that biosimilars of adalimumab (SB5, AVT02), etanercept, and infliximab (CT-P13) exhibit PK equivalence and comparable efficacy to originator biologics in the pediatric population.^{47–50} Although high discontinuation rates attributed to subjective factors rather than objective disease worsening or adverse reactions have been observed, biosimilars provide a cost-effective alternative without compromising PK targets.⁵¹ Given their established PK equivalence, biosimilars share the identical dosing regimens and TDM-guided adjustment strategies as their originator.

Model-Informed Therapeutic Drug Monitoring and Dose Optimization

While the models in Section 2 predict drug exposure, it is required to link these predictions to therapeutic endpoints in actual clinical practice. Exposure-response (E-R) analysis establishes this target window by balancing efficacy against dose-dependent toxicity ([Table S2](#)). Clinically, C_{trough} is preferred for TDM rather than area under the curve (AUC) or maximum concentration (C_{max}). While AUC best reflects overall exposure, the dense blood sampling required is impractical for children. Conversely, C_{max} primarily indicates acute toxicity rather than sustained efficacy. For mAbs with long half-lives, C_{trough} reliably surrogates overall exposure. Crucially, C_{trough} represents the most vulnerable point in the dosing interval; maintaining an adequate trough level is mechanistically essential to ensure continuous target engagement.²⁰

This Section will review E-R relationships established in pediatrics, discuss the impact of immunogenicity, and highlight inter-individual PK variability that necessitate model-informed TDM approaches to optimize therapy.

Exposure-Response Relationships and TDM Targets

To provide a quantitative basis for TDM, we summarized data from 14 representative studies in [Table S2](#), which demonstrated the key target C_{trough} and associated outcomes for major mAbs.

Substantial evidence confirms a clear E-R relationship for key mAbs across PIMDs.^{20,52–54} As summarized in [Table S2](#), multiple studies on infliximab and adalimumab in IBD have established target C_{trough} . However, the specific targets vary significantly based on the clinical endpoint (eg, clinical remission vs endoscopic healing) and timing (induction vs maintenance).^{53–58} Similar E-R relationship is also observed with tocilizumab, etanercept, and canakinumab, which require sufficient exposure to meet the American College of Rheumatology pediatric response criteria for JIA.^{59–62} However, the significant differences in target concentrations reported in [Table S2](#) highlight a challenge in pediatric TDM. Due to differences in the diverse study designs, patient populations, the disease severity, and the detection methods, the optimal concentration reported varies. To overcome this limitation, objective biomarkers are increasingly utilized. These include markers like fecal calprotectin (F-CPT), as well as specific mechanistic mediators such as CXCL9 and ferritin.^{63,64} Critically, these biomarkers often serve as the crucial PD endpoints for the PK/PD models discussed in Section 2, bridging the gap between drug concentration and target engagement. Given the absence of a universal gold standard, these biomarker-based models become essential to individualize TDM targets.

Ultimately, the clinical necessity for individualized regimens is driven by the high failure rate of standard dosing in pediatric patients. Evidence summarized in [Table S2](#) illustrates that a large proportion of children cannot achieve the target concentration through standard dosing. Conversely, proactive TDM demonstrates clear benefits, significantly reducing the need for biologic switching ($P < 0.001$) while sustaining high rates of long-term remission and mucosal healing in pediatric CD.⁶⁵ Therefore, TDM enables clinicians to early identify inadequate exposure and guide dose optimization, where model-informed approaches can provide the solution.

Impact of Immunogenicity and Dynamic Clinical Factors on Pharmacokinetics

A primary factor disrupting the established E-R relationship and driving treatment failure is immunogenicity, or the development of anti-drug antibodies (ADAs).⁶⁶ The mechanism is that ADAs markedly accelerate PK clearance by forming immune complexes with the drug.⁶⁷ This process can lead to subtherapeutic or even undetectable drug concentrations, directly compromising PD goals, such as B-cell depletion.⁶⁸

The development of ADAs has two severe clinical consequences. The first is a loss of efficacy, as the presence of ADAs strongly predicts poor treatment response and a substantially higher risk of drug discontinuation.⁶⁹ This risk is highest when ADAs co-exist with low drug trough levels.⁵³ The second is an increase in safety risks; high-titer ADAs, for example, have been linked to a more than 16-fold increase in infusion-related reactions.⁷⁰

This challenge is particularly relevant in pediatric populations. Multiple studies confirm that children and adolescents are a high-risk group, demonstrating a higher incidence of ADAs compared to adults.⁷¹ Factors such as younger age at treatment initiation have been identified as significant predictors for ADA development.⁷² A plausible hypothesis to explain this difference was that anti-tumor necrosis factor (anti-TNF) agents induced an alteration of the children's immune system toward an easier onset of hypersensitivity. Moreover, the impact of the developing immune system on immunogenicity of ADAs and higher disease activity in younger patients could also explain the higher prevalence of ADAs.^{71,72}

In addition to patient characteristics, drug exposure status is also an important risk factor. A study by Kim et al identified infliximab C_{trough} as the only factor correlated with ADA formation. They determined that a $C_{trough} < 1.88 \mu\text{g/mL}$ was the optimal cutoff for predicting ADA formation. The above research reveals the relationship between drug exposure and immunogenicity represents a complex interplay driven by multifactorial PK variability.⁷³ This complex interplay exposes the inadequacy of standard dosing. Specifically, immunogenicity serves as a primary driver of non-linear variability in drug PK. Standard weight-based dosing fails to account for these dynamic shifts, revealing the urgent need for individualized therapeutic strategies.

To capture these dynamic factors, traditional TDM is evolving towards model-informed TDM. Instead of treating immunogenicity as a simple binary status or single titer value, advanced PopPK models incorporate the timing of ADA emergence and longitudinal titer changes as time-varying covariates.^{74,75} For instance, time-varying clearance models for infliximab have successfully captured the progressive impact of dynamic ADA fluctuations in pediatric cohorts.⁷⁶

Beyond immunogenicity, dynamic clinical factors also significantly contribute to the high inter-individual PK variability in pediatric patients. While initial dosing regimens account for standard metrics like body weight, drug clearance during the maintenance phase is continuously altered by disease-driven variables. As summarized in [Table S3](#), these variables include fluctuating inflammatory burdens (eg, elevated CRP and ESR, or low albumin), shifts in disease activity, and the use of concomitant immunomodulators like methotrexate (MTX).^{19–21,77,78} When a patient experiences a loss of response, it is clinically difficult to ascertain whether the treatment failure is driven by these dynamic factors causing rapid drug clearance, or by the emergence of ADAs.

To resolve this diagnostic uncertainty, a systematic clinical decision framework is essential. The integration of TDM, immunogenicity analysis, and stratified intervention forms a clear clinical decision pathway. As illustrated in [Figure 2](#), the presence and titer of ADAs are critical determinants in distinguishing between different causes of treatment failure and guiding subsequent therapeutic actions. The practical application of this pathway, including dose escalation, reduction, or switching, is discussed in the following section.

Practical Decision Pathways for TDM-Guided Dose Adjustment

Achieving individualized therapy requires dynamic dose adjustments based on clinical response and biomarkers. As outlined in the conceptual framework in [Figure 2](#), TDM of C_{trough} and ADAs should be considered for patients with initial non-response or secondary loss of response. The analysis of these results leads to distinct optimization strategies.⁷⁹ However, it is critical to acknowledge that the strength of evidence supporting these specific interventions varies across different mAbs and indications.

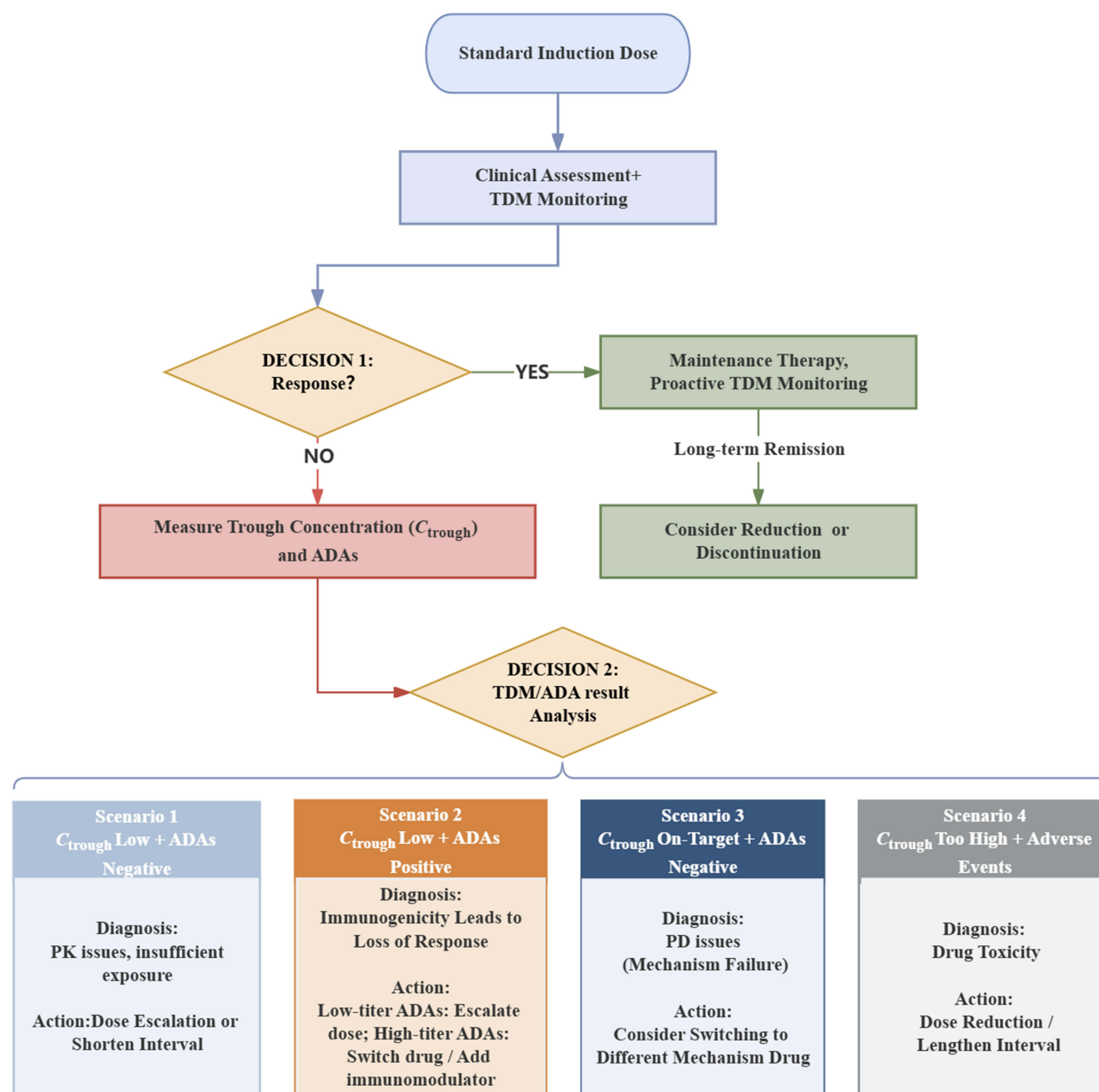


Figure 2 A Proposed Conceptual Decision Framework for TDM-Guided mAb Dose Optimization and Immunogenicity Management in Pediatrics.

Scenario 1: Subtherapeutic Exposure

This scenario suggests insufficient exposure due to PK issues rather than immunogenicity. A common strategy involves dose escalation or interval shortening.

Clinical evidence strongly supports this approach. Studies across pediatric IBD and JIA have demonstrated a clear dose-response relationship, where escalating doses (eg, infliximab to 7.5–10 mg/kg or etanercept to ≥ 0.8 mg/kg/week) or shortening intervals correlate with higher remission.^{80,81} TDM data is crucial for guiding this decision; for instance, post-induction infliximab *C_{trough}* levels have been identified as a key predictor of long-term remission.⁵³ High-dose strategies have also proven effective in refractory conditions, such as macrophage activation syndrome (MAS) or Kawasaki disease.^{82,83}

Scenario 2: Immunogenicity-Driven Loss of Response

As illustrated in [Figure 2](#), TDM-guided interventions are critical. For low-titer ADAs, dose escalation may restore clinical response. For instance, Cohen et al reported that 54% of patients with low antibody levels successfully converted to ADA-negative status after dose optimization.⁸⁴ Conversely, high-titer ADAs are less likely to be overcome. In these cases, switching to a different biologic or adding a concomitant immunomodulator is often recommended. Studies support this titer-based approach, showing that patients with low-titer ADAs have higher long-term drug survival (73%) compared to those with high-titer ADAs (50%).⁸⁵

Scenario 3 and 4: Mechanism Failure and Drug Toxicity

The TDM pathway also identifies other causes of treatment failure. If a patient has on-target drug levels (C_{trough} On-Target) but no response (and is ADA-negative), this suggests a PD issue or mechanism failure. In this case, dose escalation is futile, and the appropriate action is to switch to a drug with a different mechanism of action.

Conversely, if TDM reveals supratherapeutic levels (C_{trough} Too High), especially in the presence of adverse events, this indicates drug toxicity. This is a common challenge, as higher exposure correlates with safety risks such as infusion reactions (infliximab C_{trough} levels ≥ 20 $\mu\text{g/mL}$) or neutropenia (tocilizumab).^{86,87} The clear action is dose reduction or lengthening the interval.

Dose Reduction and Discontinuation in Remission

The TDM pathway also applies to patients who achieve long-term remission ([Figure 2](#), “YES” branch). In this cohort, proactive TDM guides dose tapering or discontinuation.

The primary goal of reduction is to minimize the long-term toxicity and burden of therapy, such as reducing concomitant corticosteroid or conventional synthetic disease-modifying antirheumatic drug (csDMARD) use.⁸⁸ However, tapering biologics carries a substantial risk of disease relapse. Clinical studies report high relapse rates (eg, over 50% in JIA within 24 months) after biologic disease-modifying antirheumatic drug (bDMARD) reduction or discontinuation.⁸⁹ This highlights the need for model-informed discontinuation strategies. Research is focused on identifying optimal tapering regimens and risk stratification. Key strategies include optimizing withdrawal sequencing (eg, discontinuing MTX before the TNF-inhibitor) and using biomarkers (eg, serum MRP8/14) to predict relapse risk.^{90,91} However, developing validated protocols for discontinuation remains a critical challenge.

Integrated Clinical Practice Recommendations

Achieving precision medicine requires summarizing the strategies discussed throughout this Section. To bridge the gap between complex PK evidence and daily practice, we have integrated the recommended dosing regimens, TDM targets, adjustment strategies, and critical covariates for major mAbs into [Table S4](#), providing a practical reference for clinicians.

Conclusions and Outlook

In conclusion, standard dosing regimens for PIMD often fail to account for the patient variability in drug disposition and target-mediated clearance. Our review highlights two core findings. First, while PopPK and PBPK models are essential for dose optimization, their accuracy still depends on the specific context and requires external validation. Second, proactive TDM in real-world practice is indispensable for mitigating subtherapeutic exposure and immunogenicity.

To overcome current bottlenecks, research in the coming years should prioritize several key areas. First, it is crucial to develop and apply next-generation mechanistic models. These models need to go beyond traditional covariates and achieve this by integrating pharmacogenomic markers, target biology, and complex drug interactions. Artificial intelligence (AI) is also a new type of data-driven prediction engine that can learn complex nonlinear relationships from high-dimensional data to predict PK/PD outcomes and complement traditional models. To successfully integrate traditional modeling with AI, we recommend establishing large-scale data registries and proactively accumulating high-dimensional EHR data. Beyond standard demographic factors, it is crucial to continuously accumulate structured data such as longitudinal pharmacokinetic profiles (eg, drug C_{trough} and quantitative anti-drug antibody titers), multi-omics data, genomic markers, dynamic immune biomarkers, comorbidities, and specific phenotypic responses across different



Figure 3 Future Perspectives of Precision Medicine with Monoclonal Antibodies in Pediatrics.

developmental stages.⁹² Ultimately, the value of predictive models lies in their clinical applications. By integrating models into user-friendly, visual Clinical Decision Support Systems (CDSS), the complexity faced by clinicians can be reduced.

The future of pediatric precision medicine is likely to depend on quantitative pharmacology, TDM, and AI (Figure 3). Urgent clinical needs have led us to develop personalized treatment plans for children and minimize their long-term treatment burden.

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Disclosure

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